

Explainable Machine Learning Framework for Early Heart Disease Prediction

Review 2 Report — Implementation & Evaluation

Machine Learning / Explainable AI Project

1. Objective

The Review 2 report presents the implementation and evaluation of the proposed Explainable Machine Learning Framework for Early Heart Disease Prediction. The main objective is to investigate whether feature selection followed by a weighted soft-voting ensemble of Logistic Regression, Random Forest, SVM, and XGBoost can provide a consistent prediction workflow while SHAP and LIME provide explanations for individual predictions.

The implementation is evaluated using the Cleveland heart-disease dataset represented by *heart_cleveland_upload.csv*. The project uses repeated stratified 5-fold cross-validation with 3 repeats to compare the individual models and the proposed ensemble.

2. Evolution of Approaches in the Literature

Generation	Representative Approach	Main Focus
1. Single Classical ML	Logistic Regression, Decision Tree	Interpretable classification using clinical features
2. Tree Ensembles	Random Forest, XGBoost	Non-linear relationships and feature interactions
3. Margin-based ML	Support Vector Machine	Learning a separating decision boundary
4. Ensemble Prediction	Voting / ensemble approaches	Combining complementary model predictions
5. Explainable AI	SHAP, LIME	Understanding feature contributions and individual predictions
6. Proposed System	Feature Selection + LR/RF/SVM/XGBoost + Weighted Soft Voting	Integrated prediction and explanation workflow with SHAP/LIME

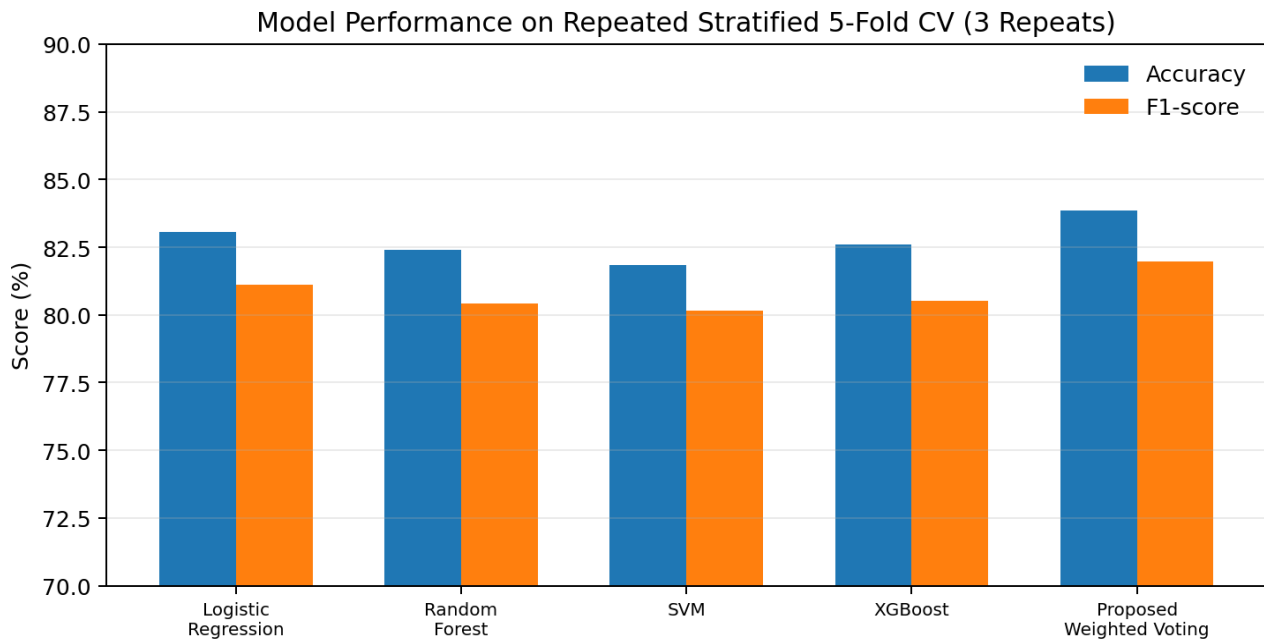
3. Comparative Analysis: Existing Models vs. Proposed System

Model / Work	Core Approach	Main Purpose	Key Features	Relation to Our Project
Logistic Regression	Linear probabilistic model	Heart-disease classification	Simple and interpretable baseline	Included as a base model
Random Forest	Multiple decision trees	Non-linear classification	Captures interactions across features	Included as a base model
SVM	Margin-based classifier	Binary classification	Learns separating decision boundary	Included as a base model
XGBoost	Gradient boosting	Non-linear classification	Models complex feature interactions	Included as a base model
Decision Tree	Single tree	Classification	Readable rule structure	Used as an additional baseline
Our Proposed Model	Feature Selection + Weighted Soft Voting	Heart-disease prediction + explanation	LR + RF + SVM + XGBoost with SHAP and LIME	Integrated ensemble and explainability framework

4. Performance Comparison Across Studies

The individual models and the proposed ensemble were evaluated using repeated stratified 5-fold cross-validation with 3 repeats. The following values are the project's reported cross-validation results.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	83.05%	84.16%	78.58%	81.11%	0.903
Random Forest	82.39%	82.62%	78.60%	80.42%	0.904
SVM	81.83%	81.02%	79.57%	80.17%	0.897
XGBoost	82.61%	83.36%	78.11%	80.51%	0.900
Decision Tree	74.08%	71.43%	73.78%	72.39%	0.741
Proposed Weighted Soft Voting	83.85%	84.58%	79.81%	81.97%	0.906



6. Key Takeaways

- **Ensemble evaluation:** The proposed weighted soft-voting model achieved 83.85% accuracy, 84.58% precision, 79.81% recall, 81.97% F1-score, and 0.906 ROC-AUC in the reported repeated cross-validation evaluation.
- **Complementary models:** Logistic Regression, Random Forest, SVM, and XGBoost contribute different modelling behaviours to the ensemble.
- **Feature selection:** The proposed pipeline applies feature selection before ensemble prediction.
- **Explainability:** SHAP and LIME are integrated to provide interpretable information about model predictions.
- **Practical deployment:** The model can be integrated into a browser-based dashboard for entering patient information and viewing the prediction and explanation.
- **Evaluation scope:** The reported values are based on the project dataset and evaluation protocol; they should not be interpreted as clinical validation on a general patient population.

7. Research Gap Addressed

Heart-disease prediction systems commonly use individual machine-learning models or tree-based ensembles, while explainability may be treated as a separate analysis step. The proposed project addresses a practical integration gap by bringing feature selection, complementary model predictions, weighted soft voting, and two explanation techniques into one end-to-end workflow.

Project flow: Patient Data → Preprocessing → Feature Selection → LR / RF / SVM / XGBoost → Weighted Soft Voting → Prediction → SHAP + LIME → Decision-Support Dashboard.

The project therefore focuses on combining predictive modelling and explainability so that a prediction is accompanied by information about influential features and the behaviour of the individual case.

8. Conclusion

The Review 2 implementation demonstrates an explainable heart-disease prediction pipeline based on feature selection, Logistic Regression, Random Forest, SVM, XGBoost, and weighted soft voting. In the reported repeated stratified 5-fold cross-validation with 3 repeats, the proposed ensemble achieved 83.85% accuracy, 84.58% precision, 79.81% recall, 81.97% F1-score, and 0.906 ROC-AUC.

The system extends prediction with SHAP and LIME explanations and can be presented through a practical dashboard. The resulting framework is a research and decision-support prototype that can be further evaluated on larger and more diverse clinical datasets and with additional validation procedures.